PHY F376: A study of hybrid classical-quantum algorithms

Week 10 (23/10/2026) - Resolving the CCSD-Seeded LUCJ Validation with Variational Optimization

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Overview of This Week's Work

Last week the CCSD-seeded LUCJ circuit failed its validation: its energy was about 60 mHa above the exact value instead of the expected 0.09 mHa. I reported this to Prof. Indrakshi, suspecting a mismatch in orbital indexing, but her reply showed that this was not the cause. This week I corrected the procedure, reproduced her noiseless benchmarks, and prepared the hardware run, which is still in progress.

What Was Wrong With the Earlier Validation

The amplitudes are consistent with the Hamiltonian. The issue was that converting the CCSD amplitudes into LUCJ parameters only gives an *initial guess*, and the circuit built from it is not expected to keep CCSD accuracy. The option `optimize=True` in from_t_amplitudes only improves how well the circuit matches the amplitudes, and does not lower the energy. A second step is required: a genuine variational minimization, in which the energy of the circuit's state is minimized directly over all of its parameters. I had treated the seeded circuit as a finished result, so the large error was expected and was not a bug.

Implementing the Corrected Procedure

I followed the supplied steps in order. The circuit was seeded from the supplied t_1, t_2 amplitudes using the local interaction pattern she specified, and the parameters were then minimized against the same Hamiltonian using a gradient-based optimizer, restarting from the seed plus small random perturbations and keeping the best result. For speed the energy was evaluated in the 36-state subspace with two spin-up and two spin-down electrons, and the final parameters were re-checked on the actual qubit circuit. Two checks are now automatic: the energy from the circuit must equal the subspace energy, which confirms the qubit ordering, and all probability must lie in the correct particle-number sector. Both checks passed.

Noiseless Results

For both $n_{\text{reps}} = 3$ and 4, the raw seeded circuit had an error of about 38.8 mHa, which fell to about 0.37 mHa after minimization. This agrees with the expected value of about 0.4 mHa for $n_{\text{reps}} = 4$ and is better than the value of about 5 mHa for $n_{\text{reps}} = 3$. Running the unchanged SQD post-processing on 10,000 simulated shots, the error fell from about 11 mHa with 5 determinants to about 2.4 mHa with 10, and reached 0.3647 mHa from 15 determinants onward, well inside chemical accuracy (1.6 mHa) and matching the expected 20-determinant result. It stops improving because the ideal state only produced 12 distinct bitstrings, so SQD cannot go beyond what was sampled. Using $n_{\text{reps}} = 4$ gave no accuracy gain over 3, so the shallower circuit is the better choice for hardware.

Hardware Preparation and Status

The workflow now transpiles the circuit for the chosen IBM backend and records its two-qubit depth and physical qubit layout, then submits 10,000 shots and runs the determinant sweep with mean and spread over repeated sampling seeds. On a stand-in heavy-hex backend the two-qubit depth was about 80 for $n_{\text{reps}} = 3$, higher than the value of about 48, so the real figures will be compared once available. The hardware run itself has not finished: my Colab session repeatedly disconnected during the long noiseless stage. I therefore saved the optimized parameters and added a short hardware-only path that prints the job identifier immediately after submission, so that results can be retrieved later even if the session drops.

Concluding Remarks

This week's outcome is that the earlier validation failure is explained and the noiseless benchmarks are reproduced, giving confidence in the Hamiltonian, ordering, and SQD pipeline before committing QPU time. The next step is to complete the hardware run for $n_{\text{reps}} = 3$ and report its valid-shot fraction and SQD error, followed by the H_2 and triplet O_2 benchmarks and the reaction-energy calculation.